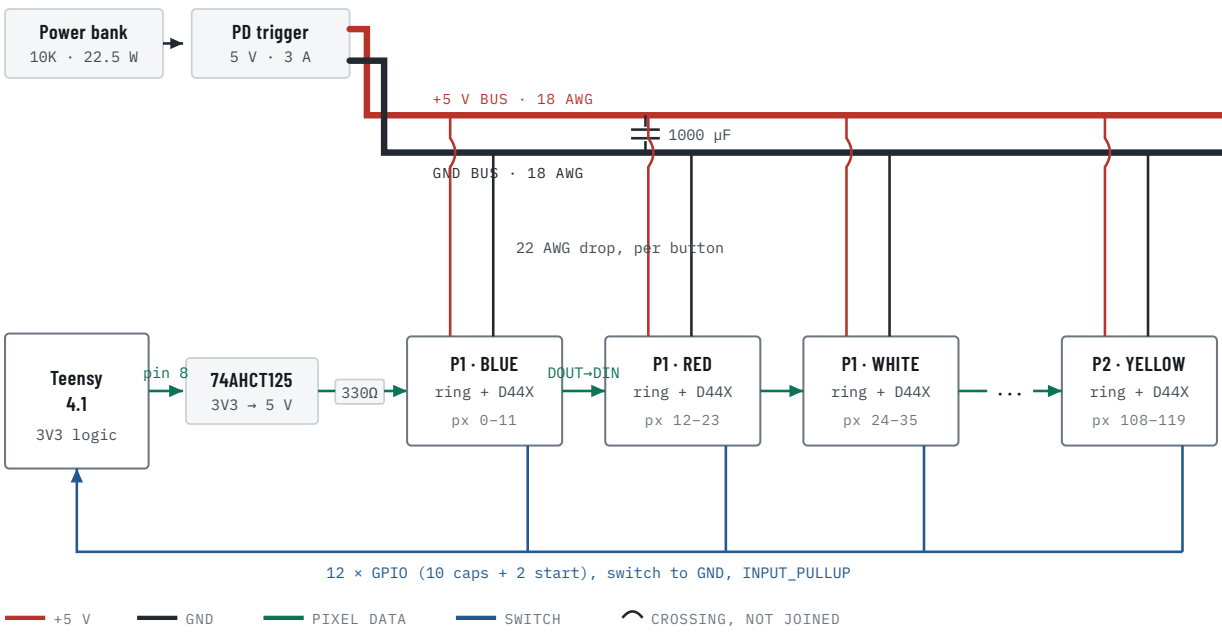
Mole Table Wiring

Ten 60 mm dome buttons carrying a 12-pixel NeoPixel ring and a Cherry D44X switch, plus two unlit start buttons, back to one Teensy 4.1. Pin assignments are mirrored from `firmware/WhackAMole/Config.h`.

> ### The one rule
>
> Power does not daisy-chain through the buttons. Only the data line does. A 12-pixel ring pulls **720 mA** at full white, so chaining 5 V through the button connectors would put **7.2 A** through the first one. Run a separate 5 V/GND bus and tap each button off it.

System topology

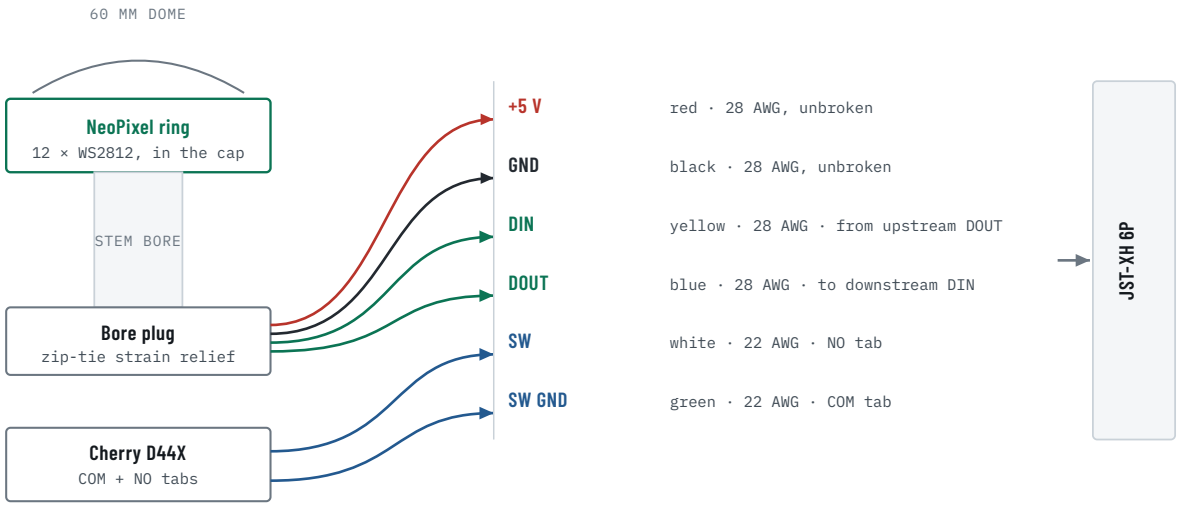
10 lit caps · 120 pixels



Two independent networks. Power fans out from a shared bus so no connector carries more than one ring's current; data runs series through every ring in turn, which is why one unplugged cap goes dark for everything downstream of it. The two start buttons sit outside both networks — switch only.

18 AWG bus, 22 AWG drops 26 AWG point to point 22 AWG returns

Inside one cap 6 conductors out



Keep the two grounds separate all the way back. A ring at full white returns 720 mA; sharing that return with the switch puts an IR offset across the input the Teensy is trying to read.

Cap pigtails one harness per button, cap to PCB

The button board is a PCB mounted **centrally under the table**, and every cap reaches it directly: one six-conductor pigtail per button, terminated in a `XHP-6` plug, long enough to run from its dome to the board with no intermediate joint. There is no per-cap junction and no bus to tap — ten identical harnesses, each one plugged into its own connector.

That is the whole reason the PCB is worth building. The stripboard version needed a 5 V/GND bus soldered across twenty strips with a stripped window at each of ten tap points; here the board distributes power in copper and the harness carries it.

Conductor colours

The same six colours on every cap, in the order the connector expects. **These are conductor functions, not cap identities** — the yellow wire is pixel data in all ten harnesses, including the ones inside the red and the green domes.

PIN	SIGNAL	COLOUR	GAUGE	TERMINATES AT
1	+5 V	red	28 AWG	Ring `5V` pad
2	DIN	yellow	28 AWG	Ring `DI` pad
3	GND	black	28 AWG	Ring `GND` pad
4	DOUT	blue	28 AWG	Ring `DO` pad
5	SW	white	22 AWG	D44X **NO** tab, via 0.187" receptacle
6	SW GND	green	22 AWG	D44X **COM** tab, via 0.187" receptacle

Two gauges in one harness, and the reason is mechanical both times. Pins 5 and 6 are 22 AWG because that is the only gauge inside both the spade receptacle's 22–18 AWG range and the JST contact's 22–28. Pins 1–4 are 28 AWG because nothing forces them to be thicker: a cap draws about **180 mA** at the configured brightness, which is **38 mV** round trip over half a metre, so the ring sees 4.96 V. Raise the brightness to full white and it is 153 mV — still 4.85 V, and still nowhere near anything a WS2812B reacts to. Thermally it is not close either: 180 mA against roughly 1.4 A for 28 AWG in free air.

One conductor from the ring pad to the connector, with no joint in between. An earlier revision ran 30 AWG inside the cap and spliced to a heavier pigtail, on the grounds that 30 AWG flexes best across the plunger travel. That is gone, and the reason it had to go is that **30 AWG is below the XH contact's 22–28 range** — it could never be crimped, so the splice existed only to get back into spec. Running 28 AWG the whole way is in spec at both ends and removes four solder joints per cap: forty across the table, every one of them buried in a stem bore where it cannot be inspected.

If you are buying wire rather than using 28 AWG you already have, **26 AWG** is the better pick — mid-range for the contact instead of at its thin edge, and 1.65× the copper. Either is fine. Use **stranded silicone** either way: at these gauges the insulation decides how well it survives the flexing, and PVC stiffens.

> ### With no splice, the bore-plug strain relief is load-bearing
>
> The splice used to absorb any pull on the harness. Without it, the far end of a half-metre run is a 28 AWG conductor soldered to a small pad on the ring, and **every tug on the pigtail arrives at that pad**. The zip tie at the bore plug stops being good practice and becomes the part that keeps the ring alive.
>
> Cinch it on the **bundle jacket, never on individual conductors** — a tie pulled down onto bare 28 AWG insulation is a slow wire stripper. Leave a 15 mm service loop on the ring side of the tie, so the plunger travel is taken up by slack and not by the solder joints.

Pins 1–4 and pins 5–6 need different crimp nests

28 AWG and 22 AWG sit at opposite ends of what an `SXH-001T-P0.6` accepts, and on an SN-28B those are two different nests. Crimp the four 28s, change nest, then the two 22s. The 22 nest closed onto a 28 AWG conductor gives a joint that passes a gentle tug today and lets go in six months — tug-test every contact *before* it goes into the housing, while it can still be cut off and redone.

Where the series resistors go

A WS2812 chain wants a series resistor of a few hundred ohms at each data input. There are ten hops in this chain and **only the first one passes through the controller board**, which decides where the resistors can physically live.

HOP	DRIVEN BY	RESISTOR SITS
Into cap 1	74AHCT125 `1Y` on the controller board	**On the stripboard** — Jumper 4, between `1Y` and ribbon position 5. Already in the cut-and-jumper tables.
Into caps 2–10	the last WS2812 *inside the previous cap*	**At the receiving ring's `DI` pad**, inside that cap. Nine of them.

Nine of these cannot go on the controller board, and it is not a matter of taste. The data for hops 2–10 leaves one cap's ring `DO`, crosses the PCB on `DATA_J1_J2` and its siblings, and enters the next cap's `DI`. The controller board is not in that path at any point, so a resistor there has no electrical relationship to the signal. **Cap 1 gets none of its own** — it already sits behind the stripboard's resistor, and two in series would put 660 Ω on that line, past the useful range.

Fit them as part of the ring wiring: one leg into the `DI` pad, the yellow 28 AWG onto the other leg, heatshrink over the body. That adds no joints — the wire was going onto that pad regardless, the resistor just lands in the middle of it. Keep the leg lengths short and let the bore plug support the body.

These nine are insurance, not a requirement

A pixel chain is not a passive bus — every WS2812 re-clocks and re-drives the signal, so the metre-long hops this star topology creates are driven by an active repeater at each end and will usually work bare. The reason to fit them during assembly is that retrofitting one means opening a cap. If you would rather not, build the chain, run `LED WALK`, and add a resistor only to a cap that sparkles or shows the wrong colour.

Length

One measurement decides the wire order, and it is the only number here that depends on your table rather than on the parts. Measure from the **farthest** cap's stem bore to where the PCB will sit, follow the route the

harness will actually take — down the leg, along a rail, not the straight line — then add 15 % for dressing and 80 mm of service loop at the board so a connector can be unplugged without tension on it.

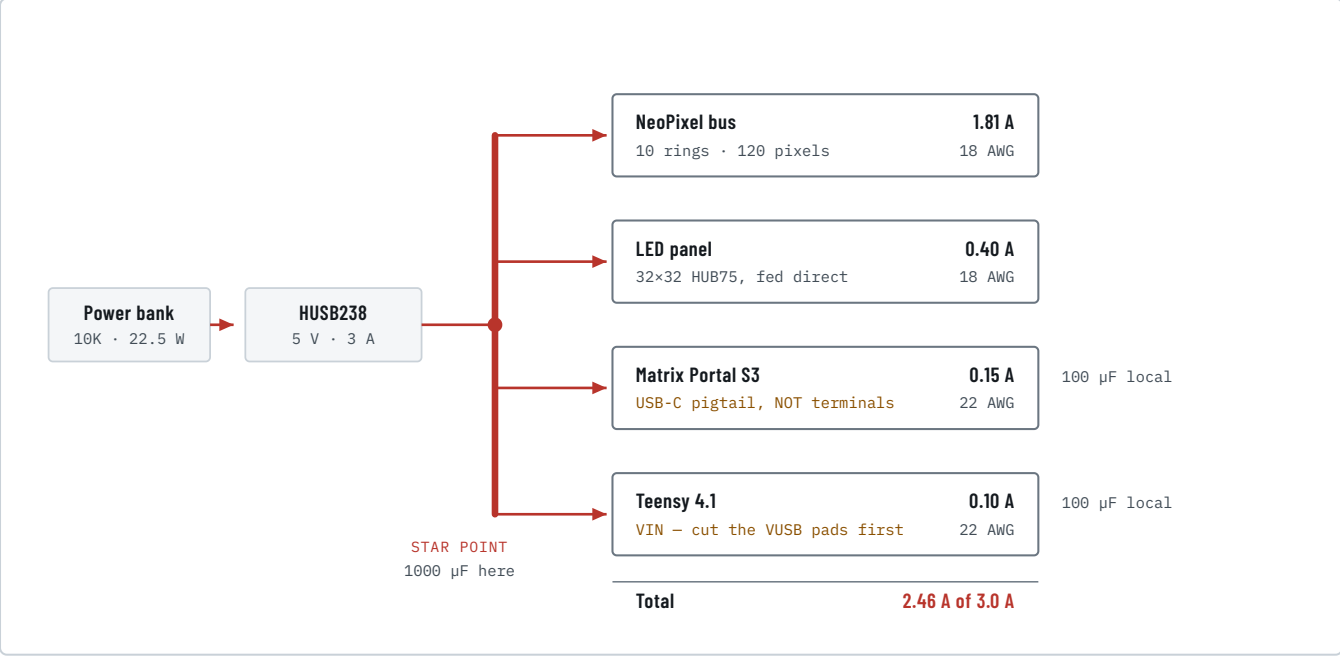
Cut **all ten to that same length**. Harnesses that differ per position cannot be swapped when one fails, and the saved wire is worth less than the ability to move a cap. For a board at the centre of a two-player table, expect 400–600 mm each, so budget roughly 6 m of each of the four 28 AWG ring colours and 7 m of each of the two 22 AWG switch colours — the switch pair serves the two start buttons as well. Order 30 ft a colour and stop thinking about it.

Start buttons

Two conductors, no ring, no power: `XHP-2` into `J11` and `J12`. **White** to the **NO** tab and **green** to **COM**, both 22 AWG — the same two colours and the same gauge as pins 5 and 6 of a cap harness, because it is the same circuit. Nothing on the start buttons is lit.

Power distribution `one supply, four loads`

All four loads run from the single 5 V contract. What matters is that they are fed as a **star from one point**, not chained one board to the next. The rings switch current fast; share a thin feed with them and those spikes sag the rail far enough to reset a microcontroller, which presents as random crashes rather than as a power problem.



Four pairs from one point, not a chain. The two annotated loads each need a board modification before they can take bus power safely.

Both microcontrollers need work before they take bus power

Teensy 4.1 — cut the VUSB/VIN pads. They are bridged from the factory, so feeding `VIN` from the bus while USB is plugged into a computer back-feeds the bank into the computer's USB port. This is not optional here: debug mode reports over USB serial, so both will routinely be connected. Cut the two rectangular pads on the underside and confirm with a meter that they are open. It is reversible with solder — and note that afterwards USB no longer powers the board, so it needs the bus even to program.

Matrix Portal S3 — its screw terminals are outputs, not inputs. They are fed from its USB-C port and exist to pass power out to the panel. Adafruit discourages back-feeding them because plugging USB in at the same time can damage the board. Power it through its USB-C instead, from a pigtail carrying 5 V and ground off the bus; the board has its own CC pulldowns, so a dumb source is fine. Unplug that pigtail when connecting a computer. Feed the panel straight from the bus, not through the Portal.

Adafruit publishes no continuous current rating for the HUSB238 breakout — only that the chip negotiates up to 5 A. Nearly 2.5 A through a small board and a 2-pin terminal block is real current, so check it for warmth on the `LED ALL` screen. If it runs hot, take VBUS through a separate high-current path and leave the HUSB238 doing nothing but the negotiation.

Current budget

worst case is a debug screen

DRAW AT 5 V	PIXELS LIT	CURRENT	NOTES
Gameplay, two moles up	24	≈ 0.30 A	Amber, not white. The normal case.
Attract mode	0–24	≈ 0.15 A	Quiescent draw is ~1 mA per pixel.
Debug `LED ALL`, white	120	1.81 A	At `kPixelBrightness = 64`. The most the rings ever pull.
Same, brightness 255	120	7.20 A	Why the ceiling exists.
32×32 LED panel	—	≈ 0.40 A	Scrolling text on black. Every pixel white would be ~2 A, which the game never displays — do not budget against that figure.
Matrix Portal S3 board	—	≈ 0.15 A	The controller itself, separate from the panel it drives.
Teensy + level shifter	—	≈ 0.10 A	
Worst case — `LED ALL` plus the rest	**120**	**≈ 2.46 A**	**12.3 W, against a 15 W supply.**

And what the supply delivers:

SUPPLY	AT 5 V	NOTES
Power bank, 5 V PD contract (`in use`)	3.0 A · 15 W	Straight onto the bus. No conversion, no converter. ~18% headroom over worst case.
Power bank, 9 V PD → XL4015	≈ 3.5 A · 20 W	The path not taken. More headroom, more parts — see the decision record below.
2× 21700 in series → XL4015	sags	Falls below the XL4015's 8 V input floor for most of the discharge.

Two things to verify, not assume

The 3 A contract has to be real. A USB-C source only offers 3 A if the sink asks properly over the CC lines. A plain breakout without the 5.1 kΩ pulldowns gets no VBUS at all on a C-to-C cable; a generic one may only be offered 500 mA. Use a trigger board configured for 5 V, and a **C-to-C cable** — a USB-A-to-C cable has no CC lines and cannot carry a PD contract.

The bank has to stay awake. Power banks shut down below roughly 50–200 mA, and attract mode is the quietest the table ever gets. An active PD contract usually prevents it, but it varies by model — leave the table idling for half an hour before an event, not during one.

Then put a meter on the 5 V rail on the `LED ALL` screen, which is the only state that reaches worst case.

Why there is no converter `decision record`

The table originally ran two 21700 cells in series into an XL4015 buck converter. That pack falls from 8.4 V to 6.0 V across a discharge and the XL4015 needs 8 V at its input, so for most of a session the converter was running below its own input floor. Fixing that is what started this change; it is worth recording why the fix deleted the converter instead of replacing the pack.

OPTION	AT 5 V	ASSESSMENT
3S 21700 pack → XL4015	≈ 4.4 A	Fixes the input floor and nothing else. Still loose cells, still a balance charger, still a pack that can be shorted under a table kids are hitting.
Power bank → 9 V PD → XL4015	≈ 3.5 A	Works, and leaves the most room. Costs a trigger, a converter, a Schottky, and ~1.5 W as heat.
Power bank → 5 V PD chosen	3.0 A	Fewest parts. The ceiling is lower and fixed.

What decided it

The current budget, once it was corrected. The first version counted the matrix panel at its all-pixels-white figure of 2.0 A and stacked that against the `LED ALL` debug screen — two worst cases that never occur together. The panel shows scrolling text on black, nearer 0.4 A. Real worst case is 2.46 A across all four loads, and gameplay is nearer 1 A, so the converter was converting power the table did not need converted.

What it buys

- Three fewer components in the power path: the converter, its Schottky, and the wiring between them.

- No conversion loss. The XL4015 is non-synchronous and runs 87–93%, so roughly 1.5 W would have gone to heat at worst case.
- The 8 V input floor stops being a constraint that can ever bite, rather than one worked around.
- A regulated 5 V rail from the bank's own converter, which is a better one than the XL4015.

What it costs

- **A hard 15 W ceiling with no room to grow.** The 9 V path would have given 20 W.
- **Audio is still to come.** A MAX98357A pulls around 0.6 A on peaks. Gameplay plus audio is ~1.6 A and fine, but `LED ALL` at full brightness with sound playing would exceed 3 A. That collision only exists in a diagnostic state, which is why it was accepted.
- **No regulation against cable drop.** Whatever the USB cable and connector lose is lost from the rail, rather than being corrected by a converter downstream of it.

How to reverse it

Change the trigger's configuration resistor from 5 V to 9 V and put the XL4015 back between it and the bus. One resistor and one board — which is the reason this was a comfortable decision to make early, before any of it is soldered down.

Chain order `must match Config.h`

The physical daisy-chain order and the pixel indices in `Config.h` are the same fact written twice. Wire them in this sequence or the wrong cap lights.

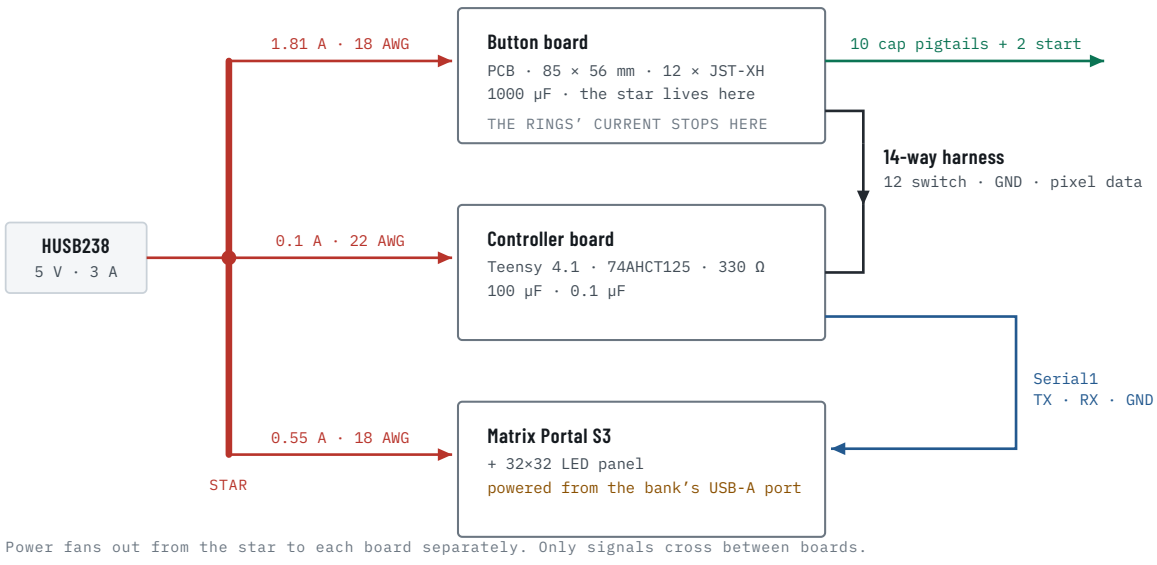
POSITION	BUTTON	PIXELS	SWITCH PIN
1–5	P1 mole caps — blue, red, white, green, yellow	0–59	2, 3, 4, 5, 9
6–10	P2 mole caps — blue, red, white, green, yellow	60–119	11, 12, 14, 15, 16
—	P1 start · P2 start (no lamp)	none	22 · 25

The start buttons are switch-only

They are plain arcade buttons with no lamp, so they contribute a switch and nothing else: two conductors, no ring, and no place in the pixel chain. `kPixelCount` is `kButtonsPerPlayer × 2 × kPixelsPerButton` — 120, not 144. Allocating pixels for them would shift nothing electrically but would leave the chain expecting two rings that are not there.

Two boards `controller and breakout`

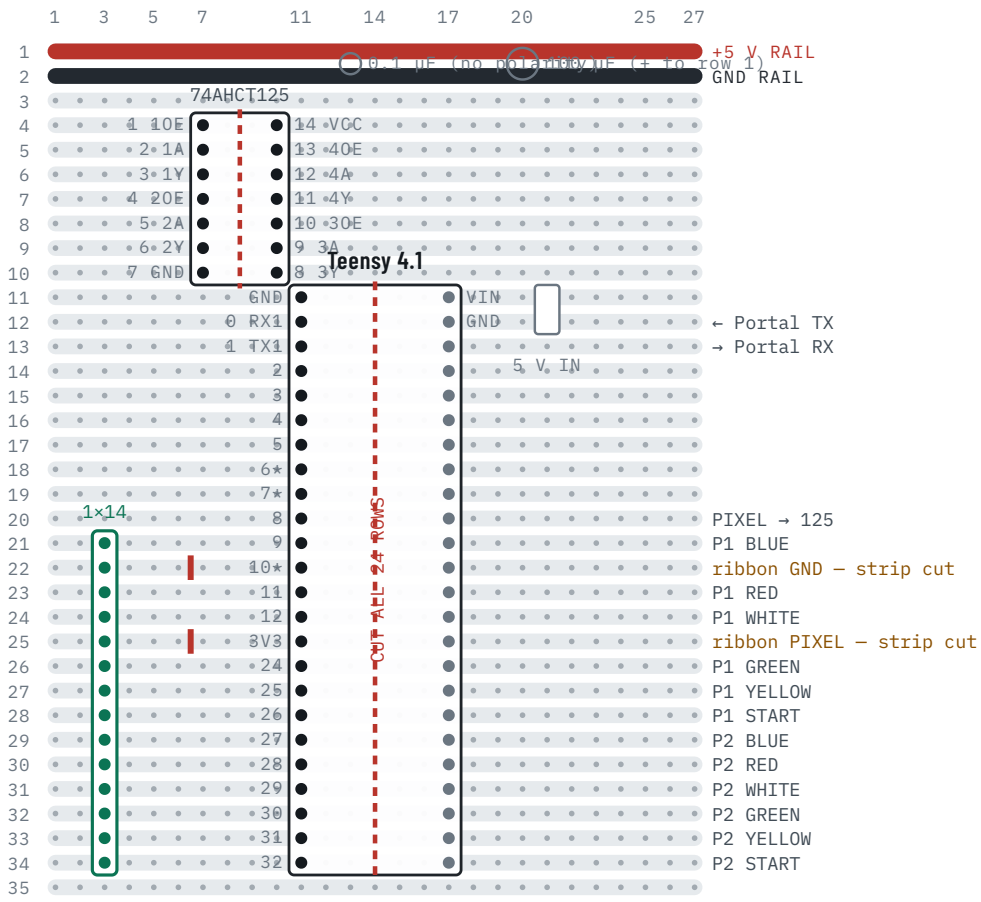
Everything does not fit on one 73 × 100 mm stripboard. The Teensy alone eats 24 of the 39 strip rows, and ten six-pin headers plus their mating plugs consume both side margins with nothing left for the level shifter, the capacitors or the power entry. Splitting it also puts the ring current where it belongs — on a board the microcontroller is not sitting on.



Three independent power pairs and one signal ribbon. The 1.81 A of ring current never enters the board the Teensy sits on, and never crosses the ribbon.

Controller board

73 × 100 mm stripboard, strips running horizontally, Teensy socketed on female headers so it can come out. Positions below are indicative — check them against your own board before cutting anything.



27 × 35 holes, strips horizontal. Dashed red is a cut down a component's centreline on every row it spans; solid red is a single strip cut isolating one row.

Cut list — do all of it before a single component goes on

Teensy centreline, rows 11–34. 24 cuts between columns 11 and 17. Without them every strip shorts a left-row pin to the right-row pin opposite it.

74AHCT125 centreline, rows 4–10. 7 cuts between columns 7 and 10. Same rule — every DIP needs it. Here the shorts would be `10E` ↔V_{CC} (which disables the one gate you are using), `1A` ↔ `40E`, and GND↔ `3Y`.

Rows 22 and 25, between columns 6 and 7. Two cuts. These sever the ribbon header's strips from the Teensy's pin 10 (reserved for audio) and its 3.3 V pin, freeing those two header positions for the ribbon's ground and pixel data. Row 25 is the one that matters: without it the pixel data line is shorted to 3.3 V.

That is **33 cuts** total. Check each with a meter before soldering — a missed cut under a soldered-down Teensy is a bad afternoon.

What the strips do for free

Twelve switch signals need no wiring at all. The ribbon header in column 3 spans rows 21–34, and on twelve of those rows the strip runs uninterrupted from the header pin straight to the Teensy pin — the strip *is* the wire. That is the whole reason the pin assignment is what it is.

ROW	TEENSY	CARRIES	ROUTING
20	8	Pixel data out	Jumper to `1A` (row 5)
21	9	P1 BLUE	Strip, no jumper
23	11	P1 RED	
24	12	P1 WHITE	
22	–	Ribbon GND	Cut from Teensy; jumper to the GND rail
25	–	Ribbon pixel in	Cut from Teensy; fed by the 330 Ω
26–34	24–32	P1 GREEN → P2 START	Strip, no jumper

Which way round the capacitors go

The 0.1 µF is ceramic and has no polarity — it goes in either way. Only the electrolytics care: the 100 µF here and the 1000 µF on the button board.

On a radial electrolytic the **stripe down the can marks the cathode**, the negative lead, and the anode is the longer lead on an untrimmed part. So on this board the striped side goes to **row 2** (GND) and the long lead to **row 1** (+5 V). Backwards, an aluminium electrolytic on 5 V will heat, vent and sometimes burst — it is one of the few mistakes on this board that makes a mess rather than just not working.

Check lead pitch before you commit a hole: radial electrolytics are usually 2.5 mm, which spans one row, but ceramics of this size are often 5.08 mm, which spans two. Leads bend — just know which you have so the cap lands on rows 1 and 2 and not rows 1 and 3.

Jumpers

Everything else is strip. Numbered so they can be ticked off and metered.

#	FROM	TO	WHY
1	row 11, right of cut	row 1	Incoming 5 V up to the rail, past the Teensy's cut.
2	row 12, right of cut	row 2	Incoming ground up to the rail.
3	row 20 (pin 8)	row 5 — `1A`	Pixel data into the level shifter.
4	row 6 — `1Y`	row 25, via 330 Ω	Shifter output to the ribbon's pixel pin.
5	row 22, left of cut	row 2	**The ribbon's ground.** Its strip is cut from the Teensy, so without this the switch returns float.
6	row 4, col 10 — V_{CC}	row 1	Level shifter supply.
7	row 10, col 7 — pin 7	row 2	Level shifter ground.
8	`1OE` r4 c7 `2A` r8 c7 `3A` r9 c10 `4A` r6 c10	row 2	To **ground**. Grounding `1OE` is what enables the one gate in use; the three `nA` pins are unused inputs being held at a level so they cannot float.
9	`2OE` r7 **c7** `3OE` r8 c10 `4OE` r5 c10	row 1	To **+5 V**. `OE` is active-low, so high disables — these three outputs go high-Z. Note `2OE` is on **column 7**, not 10.

> ### Row 7 has a pin on each column, and they are opposites
>
> `2OE` is at **row 7, column 7** and goes to +5 V. `4Y` is at **row 7, column 10** and must stay **unconnected** — it is a disabled output. They sit on the same strip row either side of the DIP's centreline cut, so "row 7" on its own is ambiguous and tying the wrong one to 5 V puts a rail on an output pin.
>
> The same shape applies at row 4: `1OE` (col 7) goes to ground while V_{CC} (col 10) goes to 5 V. Always carry the column with the row.

Outputs `2Y` (r9 c7), `3Y` (r10 c10) and `4Y` (r7 c10) stay genuinely unconnected — that is what tri-state disabled means. Jumpers 8 and 9 can each be a single wire touching its several points rather than separate links, but note that 9 spans both columns of the DIP and so crosses its centreline cut; that is fine for a wire on the component side.

Button board — the PCB

Fabricated, not cut: **85 × 56 mm, two layers**, with `GND_LED` poured on both and every track at 1.0 mm. It replaces 63 stripboard cuts and 31 hand-soldered wires with fourteen connectors and one screw terminal, and

it mounts **centrally under the table** on four M3 standoffs at 78 × 49 mm centres, so every cap harness reaches it directly.

The generator in `hardware/button-board/` is the source of truth for the netlist; the board is laid out by hand against it, and the two are diffed rather than trusted.

REF	PART	CARRIES	MATES WITH
J1–J5	`B6B-XH-A`	P1 caps — blue, red, white, green, yellow	`XHP-6` cap pigtail
J6–J10	`B6B-XH-A`	P2 caps — blue, red, white, green, yellow	`XHP-6` cap pigtail
J11	`B2B-XH-A`	P1 START — switch and switch ground	`XHP-2`
J12	`B2B-XH-A`	P2 START — switch and switch ground	`XHP-2`
J13	`B14B-XH-A`	12 switch lines, ground, pixel data	`XHP-14` — see the warning below
J14	MX126 screw terminal	+5 V in from the star, 1.81 A	bare 18 AWG, no plug
C1	1000 µF / 16 V	bulk at the star point	—
NT1	net tie	`GND_LED` to `GND_SW`, single point	—
H1–H4	M3, unplated	nothing — see below	M3 standoff

J1 to J10 are in pixel-chain order, which is not a tidy physical order. The chain runs P1 blue, red, white, green, yellow, then P2 in the same sequence, and the two rows of connectors on the board pair the players for physical tidiness only. Plug by the silkscreen reference, never by position.

The screw terminal is deliberately not a JST. J11, J12 and J14 were all 2-pin `B2B-XH-A` in the first revision, which meant the power lead physically fitted both start-button sockets — and landing +5 V on a start socket puts it onto a Teensy GPIO through the switch line. A screw terminal cannot accept an XH plug at all, so that mistake stopped being possible rather than merely unlikely. It also takes the 18 AWG feed directly, with no transition.

The mounting holes are unplated on purpose. Plated holes would bond `GND_LED` to whatever the board bolts to at four points, which is exactly the many-paths-to-ground that `NT1` exists to prevent. The screws are structure, not a conductor. Use nylon standoffs if the frame is metal and you want to stop even thinking about it.

The 14-pin JST will not fit the controller's stripboard — the cable is asymmetric

JST-XH is **2.5 mm** pitch and stripboard is **2.54 mm**. Over the thirteen gaps of a 14-pin connector that accumulates **0.52 mm**; centre it and the end pins still sit 0.26 mm out, against roughly **0.18 mm** of clearance in a 1.0 mm hole. It does not seat. The six-pin version works for the same reason it is marginal — five gaps is 0.20 mm, or ±0.10 mm centred.

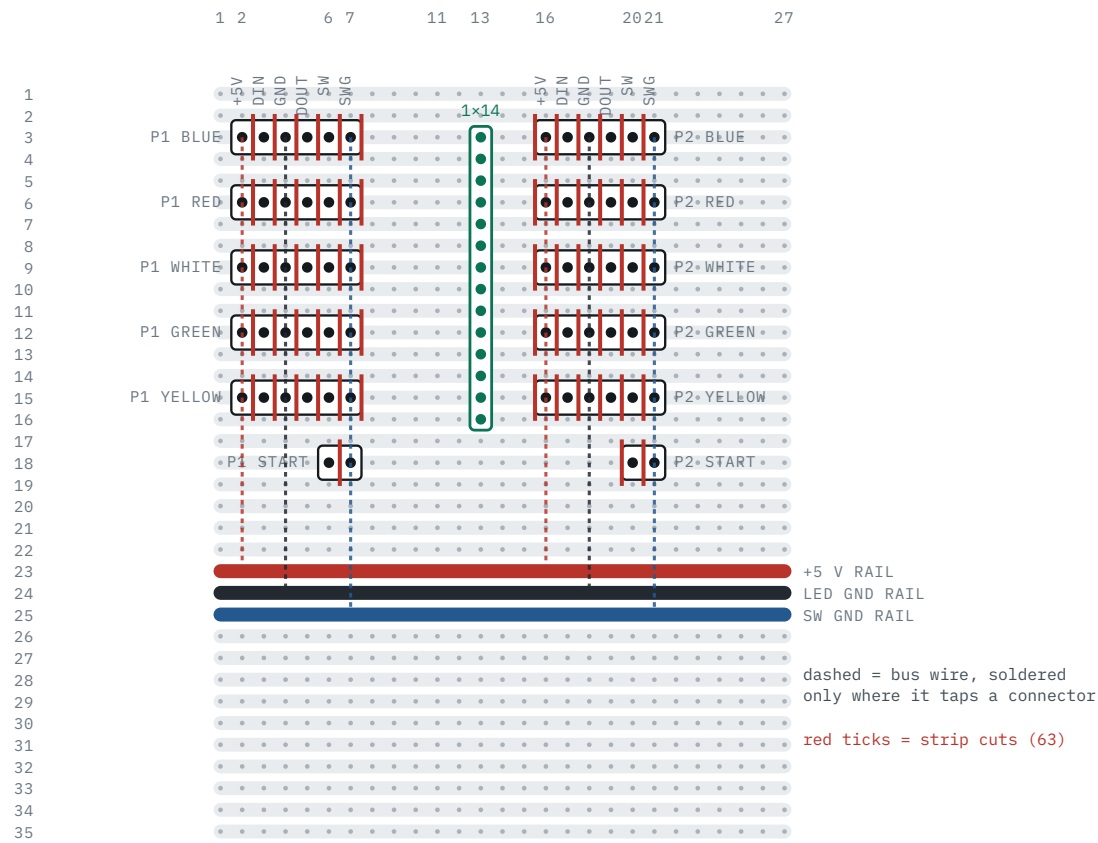
So the board-to-board cable has **a different connector on each end**: `XHP-14` into `J13` at the PCB, and a 2.54 mm 1×14 socket onto the pin header at the controller. That also means it cannot be IDC ribbon — XH needs individual crimped contacts, so this is a hand-crimped harness of fourteen conductors. Build it once, carefully, and label both ends.

Do not solve this by reaming the stripboard holes. You will tear copper off the strips the connector is supposed to be landing on.

Button board — stripboard alternative

The original, kept because it needs no fabrication and no two-week lead time. If you are building the PCB above, skip to **Wire gauges** — everything from here to there is stripboard-only, and the cut list and jumper tables do not apply to the PCB.

One dimension forces this layout. A `B6B-XH-A` is **5.75 mm deep** — 2.3 strip rows — so connectors cannot sit on adjacent rows; they need a three-row pitch. Twelve in one column wants 36 rows against the 35 available, so it has to be two columns. That also kills the tidy scheme the controller board uses, where each signal shares a strip with its Teensy pin: the ribbon's one-row pitch cannot line up with a three-row connector pitch. **Every switch signal needs a jumper here.**



Five cap pairs on rows 3, 6, 9, 12 and 15; both start buttons on row 18; the ribbon down column 13 between the blocks. Three bus rails along the bottom meet at the star.

Cuts — 63

ROWS	BETWEEN COLUMNS	×	WHY
3 6 9 12 15	2|3 3|4 4|5 5|6 6|7 7|8	30	Isolates the six pins of each P1 cap. The last also keeps SW GND off the ribbon's segment.
3 6 9 12 15	15|16 16|17 17|18 18|19 19|20	25	Same for P2. The first keeps the ribbon segment off `+5 V`. SW GND runs out to the right edge, where nothing else sits.
18	6|7 · 19|20 · 20|21	3	Separates each start button's SW from its own return.

Nothing is cut on the rows between — they carry no component, and the bus wires cross them without touching. The three rails on rows 23–25 stay uncut end to end.

Jumpers — 31 wires

WHAT	×	GAUGE	COLOUR	ROUTE
+5 V bus	2	20 AWG solid	red	Columns 2 and 16, row 3 down to the rail on row 23. Stripped and soldered at each cap's pin 1 — five taps per wire.
LED GND bus	2	20 AWG solid	black	Columns 4 and 18 down to row 24, tapped at each cap's pin 3.
SW GND bus	2	24 AWG solid	grey	Columns 7 and 21 down to row 25, tapped at every cap *and* both start buttons — twelve taps. Thin on purpose: it carries under 2 mA, and it stays a separate net so ring current never shares the switch return.
Switch signals	12	24 AWG solid	blue	Ribbon position *n* to that cap's SW pin — column 6 for P1, column 20 for P2. Diagonals of various lengths; this is what the three-row pitch costs.
Pixel chain	10	24 AWG solid	green	Ribbon position 5 into P1 BLUE's DIN, then each cap's DOUT (column 5 or 19) to the next cap's DIN (column 3 or 17). P2 YELLOW's DOUT is left open.
Ribbon ground	1	24 AWG solid	grey	Ribbon position 2 to the SW GND rail. That row has no cuts, so anywhere along it reaches.
Star tie	1	20 AWG solid	black	SW GND rail to LED GND rail, **at one point only**. This is what makes it a star — the two returns are separate everywhere else and meet here.
Power in	2	18 AWG stranded	red / black	HUSB238 to the +5 V and LED GND rails at the right-hand end. 18 AWG will not pass a stripboard hole — terminate at the edge and transition to the 20 AWG bus.

The 1000 µF sits across the +5 V and LED GND rails beside the power entry, striped lead to row 24.

> ### The chain order is not the board order
>
> Pixels run P1 blue, red, white, green, yellow, then P2 in the same sequence — down the left column and then down the right, *not* left-to-right across each row. The rows pair the two players for physical tidiness and the chain ignores that pairing completely. Follow the pixel index, not the layout.

Wire gauges

Every gauge here is chosen by a mechanical limit, not an electrical one — the currents are small enough that almost anything would carry them. The constraint that catches people first: **18 AWG solid is 1.02 mm and a**

stripboard hole is about 1.0 mm, so the gauge specified for the under-table runs physically will not pass through the board. Terminate it at a screw terminal or a board edge instead.

WHERE	GAUGE	TYPE	WHY THAT ONE
Cap pigtail — ring conductors pins 1–4, ring pad to connector	`28 AWG`	`stranded silicone`	180 mA per cap — 38 mV round trip over half a metre, against a 1.4 A free-air rating. The gauge is set by the crimp window, not the current: 22–28 is what the XH contact takes, so 28 is the thin end of legal and 26 the comfortable middle. **Silicone, and one unbroken run** — it has to flex across 3 mm of plunger travel on every press.
Cap pigtail — switch pair pins 5–6	`22 AWG`	`stranded`	The only gauge inside both the spade receptacle's 22–18 range and the JST contact's 22–28. Forced at the D44X end, so it runs the whole length.
Start buttons	`22 AWG`	`stranded`	Same circuit as a cap's pins 5 and 6, so the same gauge and the same two colours. No power, no ring.
Board to board, 14-way PCB build	`24 AWG`	`stranded`	Signals only; the ground conductor carries twelve pull-up returns, under 2 mA. Stranded and crimped, **not** IDC ribbon — the XH end needs individual contacts. See the warning above about the two ends differing.
Star to boards, star to panel	`18 AWG`	`stranded`	300–500 mm under the table, where drop accumulates. On the PCB it lands straight in the screw terminal; on stripboard, terminate at the board edge and transition to the 20 AWG bus.
Board jumpers stripboard only	`24 AWG`	`solid, insulated`	Carries signals, or the level shifter's ~20 mA. Gauge is irrelevant; **solid** is not — stranded frays going into a hole. Cat5/Cat6 offcuts are ideal: 24 AWG solid, eight colours, thin insulation.
Button board 5 V/GND bus stripboard only	`20 AWG`	`solid, insulated`	Carries up to 1.81 A. 22 AWG would do electrically (~10 mV drop over 100 mm) — 20 AWG is for robustness and for taking solder heat at ten tap points. Must be insulated: the bus crosses about twenty strips and should connect to ten. **The PCB needs none of this** — it distributes power in copper.

On the stripboard button board only: the bus is insulated wire run along the solder side, with a window stripped at each connector's +5 V and GND pad. Bare wire there would short every strip it crosses on the way past. The PCB has no bus wire at all.

Three things stripboard will do to you

The cut under the Teensy. Strips are horizontal and the Teensy mounts vertically, so every strip crossing it shorts that row's left pin to its right pin. Cut all 24 rows between the pin rows before anything else goes on the board.

Strip current, on the button board only. Stripboard copper is roughly 35 µm × 2 mm — about 29 AWG. It carries the ring current without burning, but reinforce the two bus runs there with solid wire anyway. The controller board needs none of this: its rails feed only the level shifter and its decoupling, which is tens of milliamps.

JST-XH is 2.5 mm pitch; stripboard is 2.54 mm. Over six pins that is 0.2 mm of drift against about 0.18 mm of hole clearance. It goes in, but start each connector centred so the error splits both ways rather than accumulating toward one end.

Bring-up `test before you trust`

Every test below is done with **nothing in the sockets** — no Teensy, no 74AHCT125. Installed parts add their own paths (the Teensy's protection diodes especially) and turn a clear reading into a confusing one. Put them in last.

Ribbon pinout

Header position 1 is row 21 on the controller stripboard, and position 1 at `J13` on the button board PCB. This table is the one reference both ends of the board-to-board harness are built against; it is checked against `Config.h` on every build by `docs/check-pins.py`, so it cannot quietly drift from the firmware.

POS	ROW	TEENSY	CARRIES	POS	ROW	TEENSY	CARRIES
1	21	9	P1 BLUE	8	28	26	P1 START
2	22	–	**GND**	9	29	27	P2 BLUE
3	23	11	P1 RED	10	30	28	P2 RED
4	24	12	P1 WHITE	11	31	29	P2 WHITE
5	25	–	**PIXEL DATA**	12	32	30	P2 GREEN
6	26	24	P1 GREEN	13	33	31	P2 YELLOW
7	27	25	P1 YELLOW	14	34	32	P2 START

Positions 2 and 5 are not switch lines. They land on Teensy pin 10 (reserved for audio) and on 3.3 V, so those two strips are cut away from the Teensy on the controller board and reused — position 2 for ground and

position 5 for pixel data. Everything else in the table is a switch line straight to a GPIO.

Building the 14-way harness

Fourteen conductors, 24 AWG stranded, with **a different connector on each end** for the pitch reason in the warning further up: `XHP-14` at the PCB, a 2.54 mm 1×14 socket at the controller. Crimp the XH end first — it is the one with no second chance, since a mis-crimped contact has to be cut off and the conductor is then too short.

On colour: a 14-way rainbow ribbon **repeats after ten conductors**, so positions 11–14 carry the same colours as 1–4 and colour alone cannot identify a conductor. Do not rely on it. Mark position 1 at both ends — a dab of paint or a single band of heatshrink — and verify against the table with a meter before the first power-up. The ten-colour sequence below is the usual order if you want to record what you actually built.

POS	COLOUR	CARRIES	POS	COLOUR	CARRIES
1	brown	P1 BLUE	8	grey	P1 START
2	red	**GND**	9	white	P2 BLUE
3	orange	P1 RED	10	black	P2 RED
4	yellow	P1 WHITE	11	brown (repeat)	P2 WHITE
5	green	**PIXEL DATA**	12	red (repeat)	P2 GREEN
6	blue	P1 GREEN	13	orange (repeat)	P2 YELLOW
7	violet	P1 YELLOW	14	yellow (repeat)	P2 START

> ### Two conductors in this harness are not switch lines
>
> Position 2 is ground and position 5 is pixel data. If the harness goes in reversed, pixel data lands on a GPIO configured as an input with a pull-up and ground lands on the start button line — no damage, but nothing lights and one button reads permanently pressed, which looks like a firmware fault. The pin-1 marks are what stop you chasing that for an hour.

Stage 1 — the cuts, before any component

×	PROBE A	PROBE B	EXPECT	CATCHES
24	r11–34 c11	same row, c17	**OPEN**	A missed Teensy cut. Every one, not a sample — one survivor shorts a signal to whatever is opposite.
7	r4–10 c7	same row, c10	**OPEN**	A missed DIP cut. Row 4 is the one that stops the chip working at all.
1	r22 c3	r22 c11	**OPEN**	Ribbon ground tied to the Teensy's reserved pin 10.
1	r25 c3	r25 c11	**OPEN**	Pixel data shorted to 3.3 V. The most expensive miss on the board.

Stage 2 — after soldering, sockets empty

PROBE A	PROBE B	EXPECT	CATCHES
`row 1`	`row 2`	**OPEN**	**Do this one first.** A rail-to-rail short puts the supply across a dead short at power-up.
`row N`	`row N+1`	**OPEN**	Solder bridges. Walk the dense areas: the DIP, the header, both Teensy pin columns.
`r21,23,24,26–34 c3`	`same row, c11`	SHORT	The twelve switch strips. These should already be continuous — no jumper involved.
`r11 c18`	`row 1`	SHORT	Jumper 1.
`r12 c18`	`row 2`	SHORT	Jumper 2.
`r20 c11`	`r5 c7`	SHORT	Jumper 3 — pixel data into the shifter.
`r6 c7`	`r25 c3`	**≈ 330 Ω**	Jumper 4. Not a short — the resistor is in it. A beep here means the resistor is bridged.
`r22 c3`	`row 2`	SHORT	Jumper 5 — the ribbon ground.
`r4 c10`	`row 1`	SHORT	Jumper 6 — socket V_{CC}.
`r10 c7`	`row 2`	SHORT	Jumper 7 — socket ground.
`r4c7 · r8c7 · r9c10 · r6c10`	`row 2`	SHORT	Jumper 8, each point.
`r7c7 · r8c10 · r5c10`	`row 1`	SHORT	Jumper 9, each point.
`r9c7 · r10c10 · r7c10`	`everything`	**OPEN**	The three unused outputs. If any reads to a rail, jumper 8 or 9 caught the wrong column.

Stage 3 — power, still no chips

- Apply 5 V. Current draw should be **essentially zero** — nothing is installed. Anything above a few milliamps means a short you have not found.
- Socket pin 14 (r4 c10) reads **5 V**; socket pin 7 (r10 c7) reads **0 V**. Probe the socket itself, not the strip, so the socket's own solder joints are in the measurement.

- Teensy VIN position (r11 c17) reads **5 V**; GND position (r12 c17) reads 0 V.
- Confirm the Teensy's **VUSB/VIN pads are cut** — meter across them off-board, before it goes in. If they are still joined, plugging in USB back-feeds the bank into your computer.
- **Once they are cut, USB carries data only.** The Teensy needs 5 V on `VIN` to run at all, including to be programmed — an unpowered one never enumerates, so it is simply absent from `arduino-cli board list` and looks unplugged. Power the board first, then connect USB.

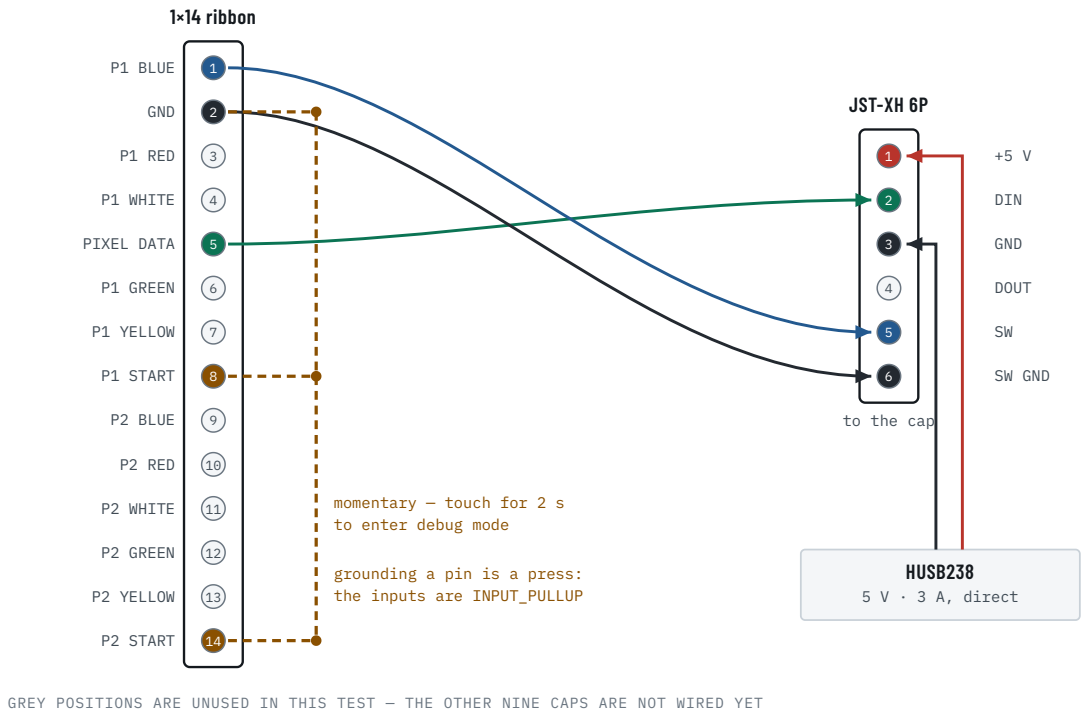
Stage 4 — chips in

Power off. Check the socket's pin-1 end matches the chip's notch. Insert the 125 and the Teensy, re-check row 1 to row 2 is still open, then power up. Steady draw should be around 100 mA — the Teensy alone. A jump means a part is in backwards.

A one-button test rig

The controller board can drive a single finished cap before the button board exists. The ribbon carries signals only, so power for the ring comes straight off the HUSB238 alongside the board's own feed. Use a spare `XHP-6` — the order has twenty.

JST PIN	SIGNAL	COMES FROM
1	+5 V	HUSB238 output, direct
2	DIN	Ribbon position **5**
3	GND	HUSB238 output, direct
4	DOUT	Nothing — end of the chain
5	SW	Ribbon position **1** (P1 BLUE)
6	SW GND	Ribbon position **2**



Three wires from the ribbon, two from the supply, one connector to the cap. The dashed amber jumper is momentary — it only exists while you are entering debug mode.

The two grounds must end up the same node — they do, because the board's ground also comes from the HUSB238. Do not power the ring from a second supply.

Getting into debug mode with one button

Debug entry holds **both** start buttons, which you do not have. The inputs are `INPUT_PULLUP`, so grounding a pin is a press: touch a wire from ribbon positions **8** and **14** to position **2** for two seconds and the firmware sees both start buttons held.

Then `LED WALK` should light your cap when it reaches P1 BLUE, and `BUTTONS` should name it when you press. That is the whole signal path proven — level shifter, data line, switch input — on one button.

Teensy 4.1 pins `from Config.h`

FUNCTION	PINS	NOTES
Display link, Serial1	`0 RX · 1 TX`	To the Matrix Portal at 115200.
Pixel data	`8`	Serial2 TX. WS2812Serial only drives certain pins; 8 sits one strip *above* the button block so the header's fourteen strips stay unbroken.
P1 buttons — B R W G Y	`9, 11, 12, 24, 25`	All `INPUT_PULLUP`, switch to ground. One unbroken block down the **left** row from pin 9 to pin 32, so the right row carries only VIN and GND.
P1 start	`26`	
P2 buttons — B R W G Y	`27, 28, 29, 30, 31`	
P2 start	`32`	
Reserved for audio	`6, 7, 10, 18, 19, 20, 21, 23`	Do not use. Pin 7 is I2S data, which is why the display moved off Serial2.
PRNG seed	`17`	Left floating deliberately.

Parts `beyond the buttons`

Part numbers are what was ordered, not the only thing that works — but the level shifter is worth matching exactly, for the reason in the note below the table.

On DigiKey the spade terminals live under *Connectors, Interconnects* → *Terminals* → *Quick Connects*, and "female" is filtered as **Receptacle**. Terminal colour is a gauge code and the same across manufacturers — **red is 22–18 AWG**, blue is 16–14, yellow is 12–10 — so a bag of blue ones is the wrong part no matter what the label said. Measure a D44X tab before ordering: arcade switches are normally 0.187" (4.8 mm) but a 6.3 mm variant exists.

PART	QTY	ORDER AS	WHY
USB-C PD sink	1	`Adafruit 5807`	HUSB238, configured for **5 V** and **3 A** by solder jumper — not I2C, which cannot be set before the Teensy has power. The current jumper ships at 1 A; leave it there and the source may cut power when the table draws 2.5 A.
Level shifter, 74**AHCT**125	2	`TI SN74AHCT125N` `DK 296-4655-5-ND`	Teensy is 3.3 V; WS2812B wants 0.7×VDD on data. 14-DIP, through-hole. See the note below.
14-pin DIP socket	2	—	On hand-soldered stripboard, turns a desoldering job into a pull.
330 Ω resistor, ¼ W	10	`any film, 5%`	**One on the controller board, nine inside caps 2–10.** The first is Jumper 4 on the stripboard, in series between the shifter's `1Y` and ribbon position 5 — that one covers the hop into cap 1, which therefore needs none of its own. The other nine go at the receiving ring's `DI` pad, because hops 2–10 never touch the controller board. See **Where the series resistors go**.
1000 µF electrolytic, **16 V**	1–2	`radial, low-ESR`	Across 5 V and ground at the star point. 16 V, not 6.3 V — convention is roughly twice the rail, so a transient has somewhere to go.
100 µF electrolytic, 16 V	3	`radial`	Local bulk at the Teensy and the Matrix Portal.
0.1 µF ceramic, 50 V	10	—	One across pins 14 and 7 of the level shifter, as close to the chip as it will sit.
0.187" × **0.020"** quick disconnect, receptacle	30	`TE 2-520181-2` `DK A27804CT-ND`	Onto the D44X tabs. 24 needed; the spares are for the ones you crimp badly. Fully insulated, so the sleeve is already on — there is no separate cover to buy at this size. The `CT` suffix is cut tape: they arrive on a carrier strip and get snipped off one at a time. 3M MNU18-187DFIK (DK 920044-01-ND) is an equivalent loose-piece part at roughly four times the price. Tab *thickness* matters as much as width — see below.

PART	QTY	ORDER AS	WHY
Crimper — insulated terminals	1	`ratcheting, 22–10 AWG`	For the spade receptacles. Insulated terminals need a matching nest profile; pliers crush the sleeve and give a joint that passes a tug test today and fails in six months. **This tool will not crimp JST contacts.**
Crimper — open barrel	1	`SN-28B or JST WC-110`	For the JST-XH contacts, which are open-barrel and need a completely different die geometry. Avoidable — see the note below.
Ribbon header, 1×14	1	`2.54 mm, through-hole`	**Controller board only** — the PCB end is a JST. Single-row on purpose: each pin lands on its own strip and aligns with a Teensy pin, where a 2×7 would short its two columns together and need a centreline cut of its own.
1×14 socket, 2.54 mm	1	`crimp or IDC`	The controller end of the board-to-board harness, onto the header above.
14-way harness wire	`~1 m`	`24 AWG stranded`	Fourteen conductors: 12 switch lines, ground, pixel data. Crimped at both ends, not IDC — see **Building the 14-way harness**.
Button board PCB	1	`85 × 56 mm, 2-layer`	Gerbers from `hardware/button-board/`. 1.6 mm, 1 oz, HASL — no option on that board matters. Allow two to three weeks.
JST-XH header, 14-pin	1	`B14B-XH-A`	`J13` on the PCB. Will **not** fit 2.54 mm stripboard — that is why the harness is asymmetric.
JST-XH housing, 14-pin	1	`XHP-14`	The PCB end of the board-to-board harness.
Screw terminal, 2-pos	1	`5 mm pitch` `MX126 / KF301 / DG126`	`J14`, the +5 V entry. Takes the 18 AWG feed directly. Deliberately not a JST so the power lead cannot be put into a start-button socket.
M3 standoff + screws	4	`nylon, 10 mm`	Mounts the PCB centrally under the table, 78 × 49 mm centres. Nylon because the holes are unplated and the ground must stay single-point.
JST-XH housing, 6-pin	10	`XHP-6` `DK 455-2218-ND`	Wire side, one per lit cap, so it unplugs for service.

PART	QTY	ORDER AS	WHY
JST-XH housing, 2-pin	2	`XHP-2`	The start buttons — switch only.
JST-XH header, 6-pin	10	`B6B-XH-A` `DK 455-B6B-XH-A-ND`	Board side, through-hole, 2.5 mm pitch. `S6B-XH-A` is the right-angle version.
JST-XH header, 2-pin	2	`B2B-XH-A`	As above.
JST-XH crimp contact	100	`SXH-001T-P0.6` `DK 455-1135TR-ND`	78 needed — 60 for the cap pigtails, 4 for the start buttons, 14 for the PCB end of the board-to-board harness. Rated **22–28 AWG**, which this build uses both ends of: the 22 AWG switch pair sits at its ceiling and the 28 AWG ring conductors at its floor. Both are in spec, but they are **different nests on the crimper** — see the warning in **Cap pigtails**. 26 AWG is the comfortable middle if you would rather not work at either edge. Arrives on a carrier strip; snip one off per crimp.
Lever nuts	6–8	`WAGO 221-415`	The star point and the 5 V/GND bus taps. Easier than a barrier strip and reusable.
USB-C pigtail, 5 V + GND	1	—	Powers the Matrix Portal through its USB-C port, since its screw terminals are outputs.
USB-C to USB-C cable	1	—	An A-to-C cable has no CC lines and cannot carry a PD contract at all.
18 AWG stranded, red + black	10 ft ea	—	The 5 V/GND bus and the panel feed.
22 AWG stranded, white + green	30 ft ea	—	The switch pair in all twelve harnesses — pins 5 and 6 of a cap, both pins of a start button. White is `SW`, green is `SW GND`, everywhere.
22 AWG stranded, red + black	15 ft ea	—	Microcontroller feeds, and the stripboard build's per-cap power drops.

PART	QTY	ORDER AS	WHY
28 AWG silicone, 4 colours red, yellow, black, blue	`30 ft ea`	—	The ring conductors of every cap pigtail, ring pad to connector in one unbroken run — so it needs the full harness length, not just the length inside the cap. Silicone for the 3 mm of plunger travel. 26 AWG is the more comfortable crimp if you are ordering fresh.

The letter that matters is the T

Order `AHCT`, never `AHC` or `HC`. The **T** means TTL input thresholds: V_{IH} is 2.0 V, so a 3.3 V signal reads reliably as HIGH while the chip runs from 5 V and drives a 5 V output. Without it, V_{IH} becomes $0.7 \times V_{CC}$ = 3.5 V and the Teensy's 3.3 V output sits *below* the threshold — which fails intermittently and with temperature rather than outright, and is miserable to chase.

A JST-XH connector is three part numbers

Housing, contacts and header are ordered separately, which is most of why they are hard to find. The housing on its own is an empty plastic shell.

That adds up to **64 crimps** across the harness, on small open-barrel contacts that ordinary crimpers mangle. Budget for an `SN-28B` or equivalent, and for an evening — with a real chance of a bad crimp hiding in the loom afterwards. Two ways around it worth pricing first: **pre-crimped leads**, which only need a contact on the plug end since the other end goes to the cap's internal wiring, and **RC balance leads** — a 5S LiPo balance connector is a 6-pin JST-XH with the wires already crimped, and they are a cheap commodity.

0.187 comes in two tab thicknesses

Width is not the whole spec. The FASTON 187 series is made for tabs of either **0.020"** (0.51 mm) or **0.032"** (0.81 mm), and the receptacles are not interchangeable. Cherry's R1 terminals — what the D44X has — are **0.187" × 0.020"**.

Fit a receptacle made for the 0.032" tab onto a 0.020" one and the spring is under-deflected: reduced contact force, and a joint that reads fine on a meter and drops out when the table gets knocked. That is the intermittent-button failure this rebuild is meant to eliminate, reintroduced at the connector. TE's application spec also notes that a receptacle once mated to a 0.032" tab may no longer mate properly with a 0.020" one, so they do not mix even as spares.

Two parts can be identical on width, gauge, insulation and colour and still be the wrong one. On DigiKey, **Mating Tab Thickness** is its own filter — set it to 0.020" before comparing prices. TE 2-520194-2 (A27806-ND) is the cheaper trap here: correct on every other line, built for the thicker tab.

Wiring the level shifter

Four gates, one used. The asymmetry to check twice: pin 1 goes to **ground** because it enables the gate you want, while pins 4, 10 and 13 go to V_{CC} because `OE` is active-low and HIGH disables. Unused inputs and unused output-enables pull opposite directions.

PIN	NAME	HOLE	CONNECT TO	WHY
1	1OE	r4 c7	**GND**	Active-low; grounding it enables the one gate in use.
2	1A	r5 c7	Teensy pin **8** (r20)	3.3 V pixel data in.
3	1Y	r6 c7	330 Ω → ribbon r25	5 V pixel data out, on to the button board.
4	2OE	r7 c7	**V_{CC}**	Gate 2 unused: output disabled, input held.
5	2A	r8 c7	GND	
6	2Y	r9 c7	—	Genuinely open. That is what tri-state disabled means.
7	GND	r10 c7	Ground	The chip's own reference. See the note below.
8	3Y	r10 c10	—	Open.
9	3A	r9 c10	GND	Gate 3 unused.
10	3OE	r8 c10	**V_{CC}**	
11	4Y	r7 c10	—	Open.
12	4A	r6 c10	GND	Gate 4 unused.
13	4OE	r5 c10	**V_{CC}**	
14	V_{CC}	r4 c10	+5 V	0.1 µF from here to pin 7, as close to the chip as it will sit.

Tie the unused inputs locally, not to the star point

Ground them to **pin 7 of this chip**, over the shortest path on the board — not with a wire running back to the bus. During pixel switching the ground bus carries amps of fast-changing current, and wire has inductance, so ground at the star point and ground at the chip differ by a couple hundred millivolts transiently. An input referenced to a distant ground can still bounce relative to its own chip's ground. Short and local is the point.

Tying the unused inputs to V_{CC} instead of GND works identically; convention is GND. What matters is only that they are not left floating — CMOS inputs that float oscillate, draw current, and inject noise into the gate you are actually using.

If the pixel chain is ever split into two, one per player, gate 2 is already sitting there: wiring the socket so enabling it means moving two wires costs nothing now.

Build order `test as you go`

01 **Do the two board modifications first**
Cut the Teensy's VUSB/VIN pads and confirm with a meter. Make the Matrix Portal's USB-C power pigtail. Both are easier on the bench than under a table, and the Teensy cut changes how you program it.

02 **Wire one cap completely**
Ring, switch, connector, all six conductors. Getting the first one right tells you the gauge and connector choices work before you repeat them nine times.

03 **Run it on the bench**
Level shifter, resistor, cap, one button. Flash the firmware with `WAM_LIGHTS_PIXELS 1` and open the debug `LED WALK` screen — one ring is enough to prove the data path.

04 **Order the PCB early**
Two to three weeks of lead time runs in parallel with everything else here. Nothing downstream of the one-cap bench test is blocked by it.

05 **Build the 14-way harness and meter it end to end**
Fourteen crimps with a different connector on each end, against the
Ribbon pinout
table. Continuity-check all fourteen and confirm no two are shorted *before* it goes near either board — it is the one part of this build that is tedious to redo in place.

06 **Populate the PCB, power it with nothing plugged in**
Screw terminal, the 1000 µF, the net tie, then the connectors. Feed 5 V into `J14` and confirm the rail at each of the ten cap connectors with the caps still unplugged. Ten minutes here beats finding a solder bridge with ten rings attached.

07 **Make the ten pigtails to one length**
All identical, all interchangeable. Label each at the connector end with its cap, because the chain order is not the board order.

08 **Add caps to the chain one at a time**
Plug them in `J1` to `J10` order. Each addition is a new failure point, and `LED WALK` names the cap it is lighting. A break shows up immediately as everything downstream going dark.

09 **Meter the rail on `LED ALL`**
The only screen that reaches worst-case current. Do this before the table sees players, not after.

10 **Verify every switch on `BUTTONS`**
Press all twelve, the two start buttons included, and confirm each decodes as the right name. This is what catches a swapped pair in the loom.